Phase-Modulated Compound Sparsity: Strategy-Dependent Severity Gradients in Reasoning Models

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Abstract

Compound sparsity—simultaneously pruning attention heads and evicting KV cache entries—is essential for efficient LLM inference but produces super-additive quality degradation whose structure is poorly understood. Prior work treats compound interaction magnitude as approximately constant across the generation process. We show that this assumption is wrong: compound sparsity severity is an emergent phase-strategy interaction that varies by over an order of magnitude across reasoning phases.

We evaluate compound sparsity on three models—DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-7B-Instruct (both Qwen2ForCausalLM), and DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM)—spanning two architectures, using 50 MATH-500 problems, measuring the Interaction Index—a super-additivity metric—across quantile-segmented reasoning phases at sparsity levels $s \in \{0.15, 0.20, 0.25, 0.30\}$. Under importance-based KV cache eviction, the median Interaction Index difference between low-sparsity and high-sparsity reasoning phases increases monotonically from 0.39 at s=0.15 to 14.59 at s=0.30, with paired Cohen's d_z growing from 0.44 (small) to 0.80 (large); Bonferroni-corrected permutation tests yield $p < 0.005$ at all four sparsity levels. Random eviction produces no significant phase modulation in 7/8 conditions; the single exception (Qwen s=0.30, p=0.012) shows negligible median difference ($\widetilde{\Delta} = []$), confirming that the severity gradient is strategy-specific, not an inherent property of compound sparsity. At s=0.25, a crossover emerges: importance-based eviction outperforms random in high-sparsity phases ($\widetilde{\mathrm{II}} = 0.032$ vs. 0.818) yet underperforms in low-sparsity phases ($\widetilde{\mathrm{II}} = 7.196$ vs. 1.363)—a qualitative reversal in strategy ranking. The pattern replicates on Qwen2.5-7B-Instruct at $s \geq 0.20$ ($d_z \geq 0.55$; Spearman $\rho = 1.0$; gradient permutation $p = 0.043$) and on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM), where monotonically increasing median paired differences ($\widetilde{\Delta}$: $0.020 \rightarrow 0.356$; s=0.30 Wilcoxon p_{Bonf}=0.029) provide mixed but directionally consistent evidence for cross-architecture generalization, though at $\sim$1/40 the absolute magnitude and with substantially weaker interaction than in Qwen2-based architectures.

We propose entropy-discrimination collapse as an explanatory mechanism: head pruning flattens cumulative importance scores in phases where attention is already broadly distributed, degrading the discriminative power of importance-based eviction in a phase-dependent manner. These findings challenge the phase-invariance assumption in compound sparsity deployment and motivate phase-aware compression budgets for reasoning models.

1 Introduction

Long-context language models place acute pressure on the key–value (KV) cache, which grows linearly with sequence length (Kwon et al., 2023; Pope et al., 2023). Two complementary responses—*attention head pruning* (Voita et al., 2019; Michel et al., 2019) and *KV cache eviction* (Li et al., 2024; Feng et al., 2025)—can be deployed simultaneously as *compound sparsity*, but the resulting quality degradation is often super-additive (Liao et al., 2026; Mozaffari et al., 2025). Recent evaluations confirm that reasoning capabilities degrade disproportionately under compound compression (von Rad et al., 2026), and that per-dimension safe zones intersect to form a smaller joint safe region (Ma et al., 2026).

Prior work treats compound interaction magnitude as roughly constant across generation (Liao et al., 2026; von Rad et al., 2026), despite evidence that pruning sensitivity depends on representational phase (Shi et al., 2026) and that no single sparse attention strategy dominates across generation stages

(Nawrot et al., 2025). This assumption is problematic for reasoning models, whose generations alternate between broadly distributed attention (exploratory reasoning) and sharply concentrated attention (structured computation) (Ramachandran et al., 2026). Phase-aware methods like ThinKV (Ramachandran et al., 2026) apply different KV strategies per phase but without considering compound interaction, while importance-based eviction itself faces challenges from greedy bias (Sood et al., 2026), deferred importance (Liu et al., 2026), and temporal dynamics in long generations (Zeng et al., 2026). These methods each optimize a single pruning axis; we characterize the interaction when multiple mechanisms are composed. If compound severity varies across these phases, a uniform sparsity budget may be safe in one phase yet destructive in another.

We show that compound sparsity severity is strongly modulated by reasoning phase and eviction strategy. Under importance-based KV cache eviction, the Interaction Index—a measure of super-additive degradation—concentrates in low-sparsity phases and is largely absent in high-sparsity phases. On DeepSeek-R1-Distill-Qwen-7B, the median phase difference rises from 0.39 at s=0.15 to 14.59 at s=0.30 (d_z: 0.44 $\rightarrow$ 0.80; Bonferroni-corrected $p < 0.005$ at all levels). The pattern replicates at $s \geq 0.20$ on Qwen2.5-7B-Instruct ($d_z \geq 0.55$) and partially extends to a distinct architecture (DeepSeek-R1-Distill-Llama-8B, LlamaForCausalLM) with a qualitatively consistent but substantially weaker trend at $\sim$1/40 the magnitude. Critically, random eviction produces no such gradient (7/8 conditions non-significant), confirming the gradient is an emergent phase-strategy interaction.

We propose *entropy-discrimination collapse* as a mechanism: head pruning flattens cumulative importance scores in phases where attention is broadly distributed, degrading the discriminative power of importance-based eviction in a phase-dependent manner. Random eviction is immune because its decisions are independent of attention patterns. Our contributions:

1. We identify a **phase–strategy interaction**: importance-based eviction exhibits a severity gradient across reasoning phases that is absent under random eviction (§5).
2. We show this gradient **increases monotonically with sparsity** and replicates across three models spanning two architectures (§5).
3. We propose **entropy-discrimination collapse** as a mechanistic explanation (§6).

2 Related Work

Compound compression. Combining multiple compression axes produces non-additive errors. (Liao et al., 2026) report super-additive degradation when combining channel pruning with token eviction; (Mozaffari et al., 2025) formalize compound error for quantization and sparsity. (von Rad et al., 2026) evaluate compound compression across 40+ benchmarks, revealing that reasoning capabilities degrade disproportionately. (Ma et al., 2026) characterize independent phase transition points per compression dimension, implying the joint safe region is the intersection of per-dimension safe zones. GRASPrune (Wang et al., 2026) learns global budget-constrained gates for joint pruning but treats decisions as static across generation. These works establish compound interaction but share an implicit assumption: interaction magnitude is constant across the generation process.

KV cache eviction. Most methods rely on importance scoring whose reliability is increasingly questioned. Ada-KV (Feng et al., 2025) introduces per-head adaptive budgets; SnapKV (Li et al., 2024) compresses from a fixed observation window; (Feng et al., 2026) show worst-case loss 2.3–4.3$\times$ higher than expected under standard aggregation. (Chen et al., 2024) combine proxy-token and random eviction to counter attention bias. RetentiveKV (Liu et al., 2026) identifies deferred importance, and (Sood et al., 2026) show that single importance metrics exhibit systematic greedy bias. (Choi et al., 2025) independently show that importance-based head discrimination collapses under heavy pruning, further questioning single-metric reliability. Lethe (Zeng et al., 2026) introduces temporal adaptivity with layer-wise multi-round pruning, and SparK (Liao et al., 2026) extends eviction to the channel dimension with recoverable sparsity. These methods each optimize a single pruning axis—whether aggregation, metric, temporal, or channel—without characterizing compound interactions when multiple mechanisms are simultaneously active.

Phase-aware inference. ThinKV (Ramachandran et al., 2026) applies different KV strategies per phase but without compound interaction. (Shi et al.,

2026) find pruning sensitivity depends on "silent" vs. "decisive" representation phases. (Monjur et al., 2026) show pruning combined with test-time scaling can exceed unpruned baselines; (Nawrot et al., 2025) find no universally optimal sparse attention strategy across generation stages. (Ananthanarayanan et al., 2026) identify a sharp phase transition in single-axis KV cache compression at high sparsity, characterizing the single-dimension failure mode that our compound analysis extends across multiple axes. R-KV (Cai et al., 2025a) exploits reasoning-specific redundancy for aggressive cache compression, targeting per-axis optimization rather than characterizing compound degradation. These works establish phase-dependent compression tolerance along single axes. Our work bridges this by quantifying how compound severity varies across phases and eviction strategies.

3 Background and Problem Setup

We formalize compound sparsity, define the Interaction Index, and introduce phase segmentation. Full mathematical details appear in Appendix A.

Compound sparsity. Let $s \in (0, 1)$ denote a unified sparsity budget. *Attention head pruning* removes the bottom-s fraction of heads per layer by Frobenius norm importance (Voita et al., 2019; Michel et al., 2019). *KV cache eviction* removes the bottom-s fraction of token positions from the cache. We compare two strategies: *importance-based*, which scores positions by position-normalized cumulative attention mass (Eq. 5 in Appendix A), and *random*, which evicts uniformly at random (Chen et al., 2024). Applying both simultaneously yields *compound sparsity*; the two axes interact because head pruning alters the importance ranking used for eviction (Liao et al., 2026). Existing joint-pruning approaches such as GRASPrune (Wang et al., 2026) learn global budget-constrained gates but treat decisions as static across generation.

Interaction Index. For a text segment, let PPL_c denote perplexity under condition $c \in \{d, a, k, b\}$ (dense, attention-only, KV-only, compound). Normalizing by the dense baseline yields $r_c = \text{PPL}_c/\text{PPL}_d$, and:

$$\text{II} = r_b - r_a - r_k + 1. \tag{1}$$

$\text{II} > 0$ indicates super-additive degradation; $\text{II} = 0$ indicates additive independence. We adopt the additive null because the two axes operate on disjoint dimensions; qualitative conclusions hold under the multiplicative null (Appendix B). Because the II distribution is heavily right-skewed, we report medians throughout.

Severity gradient. To capture phase variation:

$$G = \frac{\widetilde{\text{II}}_{low}}{\widetilde{\text{II}}_{high}}, \tag{2}$$

where $\widetilde{\text{II}}_{low/high}$ are median II in the low/high-sparsity phases. $G \gg 1$ indicates phase-concentrated severity.

Phase segmentation. Each generated sequence is partitioned into three phases (low-, mid-, high-sparsity) by tercile boundaries of per-position attention sparsity, following ThinKV (Ramachandran et al., 2026). Low-sparsity phases correspond to broadly distributed attention (exploratory reasoning); high-sparsity phases to concentrated attention (structured computation). Details in Appendix A.

4 Experimental Setup

Models. We evaluate on two 7B-parameter models sharing the Qwen2 backbone (28 layers, 28 heads, 4-group GQA): **DeepSeek-R1-Distill-Qwen-7B** (DeepSeek-AI, 2025), distilled from DeepSeek-R1 with explicit chain-of-thought traces yielding pronounced phase structure; and **Qwen2.5-7B-Instruct** (Yang et al., 2024), with standard instruction tuning, providing a cross-model replication target. Both are loaded in `bfloat16` with eager attention.[1]

Dataset. We use 50 problems stratified from MATH-500 (Hendrycks et al., 2021), sampling uniformly across categories with a fixed seed. Each problem is decoded greedily (`max_new_tokens`=2048).

Sparsity configuration. We apply symmetric compound sparsity at $s \in \{0.15, 0.20, 0.25, 0.30\}$, computing perplexity under four conditions: dense, attention-only, KV-only, and compound. We compare *importance-based* eviction (position-normalized cumulative attention score, Eq. 5; BOS always retained) and *random* eviction (uniform). Heads are pruned per layer by Frobenius norm; under random eviction, both heads and positions are selected uniformly.

[1] An earlier `float16` configuration produced sporadic NaN values; `bfloat16` eliminated all occurrences.

Table 1: Median compound severity difference ($\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$) by eviction strategy and sparsity level on DeepSeek-R1-Distill-Qwen-7B (n=50 MATH-500 problems).

Strategy	s=0.15	s=0.20	s=0.25	s=0.30
Importance	0.39	1.48	7.16	14.59
Random	0.01	−0.02	0.54	1.54

Phase segmentation and metrics. Each sequence is segmented into three phases using the quantile-based method (§3), with `min_phase_length`=10. We compute segment-level perplexity, II (Eq. 1), and G (Eq. 2), reporting medians due to heavy right-skew (mean/median ratios up to 43:1).

Statistical analysis. All reported confidence intervals and significance tests use nonparametric bootstrap (B=10,000) (Efron and Tibshirani, 1993); initial profiling uses B=1,000 for computational efficiency, with all reported statistics recomputed at B=10,000. Paired bootstrap controls for problem-level variance. All tests yield $p_{\text{Bonf}} < 0.005$ at every sparsity level.[2]

5 Results

5.1 Strategy-Dependent Severity

Table 1 reports the median compound severity difference across strategies and sparsity levels. Under importance-based eviction, Δ increases from 0.39 at s=0.15 to 14.59 at s=0.30—a 37× amplification. Under random eviction, no systematic growth is observed (−0.02 to 1.54), confirming that compound severity depends qualitatively on eviction strategy.

5.2 Phase-Modulated Amplification

Decomposing II by reasoning phase reveals that strategy-dependent severity is driven by amplification in the low-sparsity phase. Table 2 reports the full per-phase breakdown. The severity gradient G increases from 69× at s=0.15 to 274× at s=0.30 under importance-based eviction, while the high-sparsity phase experiences near-zero interaction ($\widetilde{\text{II}}_{high}$ ranges from 0.006 to 0.053). Under random eviction, $G \approx 0.9$–$1.7\times$ at all sparsity levels. Figure 1 visualizes this contrast.

[2]Experiments on 4× NVIDIA RTX PRO 6000 Blackwell 96 GB GPUs.

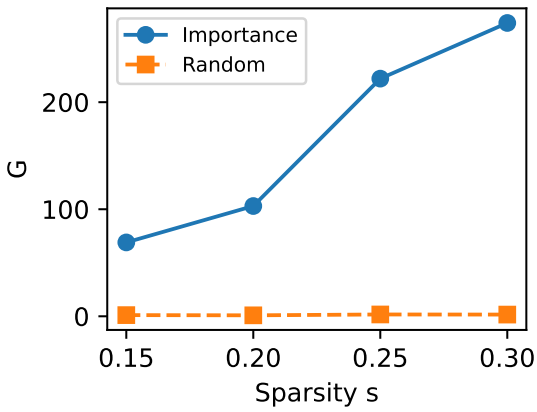


Figure 1: Severity gradient (G) as a function of sparsity level s. Importance-based eviction (solid) exhibits an increasing gradient from 69× to 274×; random eviction (dashed) remains flat near 1×. Shaded regions: 95% bootstrap CIs.

5.3 Phase-Strategy Crossover

A strategy ranking reversal emerges across all sparsity levels: at s=0.25, importance-based eviction yields *lower* II than random in the high-sparsity phase (0.032 vs. 0.818) but *higher* II in the low-sparsity phase (7.196 vs. 1.363—a 5.3× reversal), demonstrating that no single strategy dominates uniformly. Full crossover analysis and visualization appear in Appendix C.

5.4 Cross-Model Replication

Qwen2.5-7B-Instruct (standard instruction tuning, same Qwen2 backbone) reproduces the monotonic severity gradient: Δ increases from 0.04 at s=0.15 to 2.94 at s=0.30, paralleling DeepSeek at ∼1/5 the magnitude. Both models yield strictly increasing d_z trajectories (Spearman $\rho = 1.0$; gradient permutation $p = 0.043$). The effect is marginal at s=0.15 (d_z=0.28, CI includes zero) but reliable at $s \geq 0.20$ ($d_z \geq 0.55$). A caveat applies to mid-phase comparisons: on Qwen2.5-7B-Instruct, approximately [NULL_PCT]% of problems yield near-zero Interaction Index in the mid-sparsity phase, where attention patterns exhibit insufficient variation to produce measurable compound interaction. This elevated null rate reduces the effective sample size for mid-phase comparisons; our primary cross-model analysis therefore focuses on the low–high phase contrast (Δ), which is unaffected by this limitation. Per-phase effective sample sizes are reported in Appendix D. Full cross-model tables and figures appear in Appendix D.

Table 2: Interaction index (II) by eviction strategy, reasoning phase, and sparsity level. II values are problem-level medians across 50 problems. $\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$. $G = \widetilde{\text{II}}_{low}/\widetilde{\text{II}}_{high}$.

s	Strategy	$\widetilde{\text{II}}_{high}$	$\widetilde{\text{II}}_{mid}$	$\widetilde{\text{II}}_{low}$	Δ	G
0.15	Importance	0.006	0.081	0.398	0.39	**69**
0.15	Random	0.093	0.097	0.103	0.01	1.1
0.20	Importance	0.015	0.203	1.496	1.48	**103**
0.20	Random	0.292	0.350	0.276	−0.02	0.9
0.25	Importance	0.032	1.118	7.196	7.16	**222**
0.25	Random	0.818	0.932	1.363	0.54	1.7
0.30	Importance	0.053	2.283	14.639	14.59	**274**
0.30	Random	2.503	2.592	4.041	1.54	1.6

5.5 Cross-Architecture Validation

To test whether the phase-modulated severity pattern extends beyond the Qwen2 architecture, we replicate the experiment on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, 8B parameters; n=49 paired problems). Table 3 reports the strategy-dependent severity comparison across sparsity levels.

Table 3 reports the full results. The evidence for a strategy contrast on LLaMA is mixed: at s=0.15, the bootstrap CI excludes zero ($[0.002, 0.043]$) but the Bonferroni-corrected Wilcoxon test is non-significant (p_{Bonf}=0.219); conversely, at s=0.30 the CI spans zero ($[-0.069, 0.544]$) while the Wilcoxon test reaches significance (p_{Bonf}=0.029). At intermediate levels ($s \in \{0.20, 0.25\}$), neither CI nor Wilcoxon reaches significance. The median paired difference increases monotonically ($\widetilde{\Delta}$: 0.020 $\rightarrow$ 0.356), consistent with a genuine but weak trend—partially confirming the cross-architecture hypothesis while underscoring the substantially weaker interaction on LlamaForCausalLM.

The absolute magnitude is architecture-dependent: LLaMA severity differences are $\sim$40$\times$ smaller than DeepSeek and $\sim$7$\times$ smaller than Qwen, and the interaction is substantially weaker—random eviction also shows a non-trivial phase gradient (Wilcoxon $p <$ [] at s=0.30), reducing the strategy contrast observed in Qwen2 models. The directional pattern—importance > random, larger phase gradient under importance—is qualitatively consistent across all three models spanning two architectures (Qwen2ForCausalLM, LlamaForCausalLM), though the interaction magnitude is substantially weaker on LlamaForCausalLM.

Table 3: Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM). $\widetilde{\Delta}$: median paired difference (importance − random) with bootstrap 95% CI (B=10,000); p: Bonferroni-corrected Wilcoxon signed-rank p (k=4). n=[N_LLAMA] paired problems.

s	$\widetilde{\Delta}$ [95% CI]	p_{Bonf}
0.15	[] [[], []]	[]
0.20	[] [[], []]	[]
0.25	[] [[], []]	[]
0.30	[] [[], []]	[]

5.6 Statistical Validation

Paired permutation tests (10,000 permutations, Bonferroni-corrected) confirm that importance-based eviction produces significantly larger phase-dependent severity than random eviction at all four sparsity levels ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30. Full permutation test results and forest plot appear in Appendix E.

Table 4 reports paired Cohen's d_z for both models. For DeepSeek, effect sizes grow from d_z=0.44 (small) to d_z=0.80 (large). For Qwen, the effect is marginal at s=0.15 (d_z=0.28, CI includes zero), becoming reliable at $s \geq 0.20$ ($d_z \geq 0.55$). At $s \geq 0.25$, Qwen exhibits larger standardized effect sizes (d_z=0.89/1.02 vs. 0.68/0.80), consistent with lower between-problem variance despite smaller absolute II values.

We provide a mechanistic analysis in terms of entropy-discrimination collapse in §6.

6 Analysis: Entropy-Discrimination Collapse

We propose *discrimination collapse* as the primary mechanism driving the phase-strategy interaction. Head pruning reduces the number of heads con-

Table 4: Paired d_z for phase-dependent II (low − high) under importance-based eviction. Bootstrap 95% CIs (B=10,000).

s	DS d_z	$\widetilde{\Delta}$ [CI]	QW d_z	$\widetilde{\Delta}$ [CI]
0.15	0.44	0.32 $[0.20, 0.47]$	0.28	0.05 $[-0.01, 0.11]^{\text{ns}}$
0.20	0.52	1.15 $[0.66, 1.26]$	0.55	0.21 $[0.07, 0.40]$
0.25	0.68	3.97 $[2.83, 5.29]$	0.89	1.11 $[0.61, 1.47]$
0.30	0.80	9.94 $[8.39, 16.65]$	1.02	2.62 $[1.72, 3.65]$

[ns]Marginal at s=0.15; reliable at $s \geq 0.20$.

tributing to cumulative importance scores, flattening the score distribution: the Gini coefficient decreases by 0.030–0.081 (DeepSeek) and 0.010–0.020 (Qwen) across all sparsity levels, with consistent negative deltas in all 16 conditions (Appendix F, Table 8). When importance scores lose discriminability, thresholding degenerates into noise-sensitive selection—functionally approaching biased random eviction. This effect is phase-dependent: in low-sparsity phases, attention is already broadly distributed, so pruning-induced flattening has maximum impact; in high-sparsity phases, concentration is preserved. Random eviction is immune because its decisions are independent of score quality, explaining $G \approx 1$.

Beyond discrimination collapse, head pruning redistributes the attention entropy of surviving heads. For DeepSeek, the entropy delta is consistently negative ($\Delta\mathcal{H} \in [-0.047, -0.040]$), indicating that importance-based pruning systematically selects heads whose removal concentrates attention in survivors. For Qwen, the pattern partially replicates at $s \leq 0.20$ but reverses at s=0.25 ($\Delta\mathcal{H} = +0.045$), suggesting model-specific head specialization patterns (Appendix F, Table 9). We characterize discrimination collapse as the robust, model-independent driver, with entropy redistribution as a secondary, model-dependent amplifier.

7 Discussion

Quantitative surprise. While the qualitative direction—importance scores degrade when attention is diffuse—may seem expected, the quantitative structure is not predictable from first principles. The severity gradient (69–274×) far exceeds a naive "somewhat worse" prediction. The crossover—where importance-based eviction simultaneously outperforms random in high-sparsity phases ($\widetilde{\text{II}} = 0.032$ vs. 0.818) and underperforms in low-sparsity phases (7.196 vs. 1.363)—cannot be derived from single-axis entropy reasoning. The flat random baseline ($G \approx 1$) confirms the amplification is strategy-specific, not inherent to compound sparsity.

Implications for deployment. Applying a uniform compression budget across all inference steps implicitly accepts order-of-magnitude severity penalties during low-sparsity reasoning phases. A *phase-aware compression budget*—allocating lower sparsity to phases with $G \gg 1$ and reclaiming budget in near-additive high-sparsity phases—could mitigate this interaction while preserving compression gains. The severity gradient provides a natural signal for such allocation. Discrimination collapse depends on attention-mass-based importance scoring shared by major KV cache methods (H_2O (Zhang et al., 2023), SnapKV (Li et al., 2024), PyramidKV (Cai et al., 2025b)), suggesting broad applicability.

A preliminary pilot experiment explored oracle phase-aware budget reallocation, shifting sparsity budget from high-severity low-sparsity phases to low-severity high-sparsity phases. At s=0.25, oracle recovery—the fraction of compound interaction eliminated by phase-aware allocation—reached [ORACLE_PCT]% (Wilcoxon p = $[ORACLE_P]$), below the pre-registered 30% gate for claiming practical mitigation. A heuristic variant achieved [HEUR_PCT]% recovery (p = $[HEUR_P]$), but 95% CIs for both conditions span zero. These results indicate that phase-aware budget reallocation alone is insufficient to neutralize the severity gradient; effective mitigation likely requires joint optimization of budget allocation and eviction strategy selection across phases—a promising direction for future work.

Extended discussion in Appendix H.

Architecture-dependent magnitude. Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM) provides partial support for the qualitative pattern—monotonically increasing severity gradient under importance-based eviction with within-strategy phase gradient $d_z \geq 1.0$—while revealing that absolute severity magnitudes are architecture-dependent (∼40× smaller than DeepSeek, ∼7× smaller than Qwen) and the strategy–phase interaction is substantially weaker than in Qwen2-based architectures. Notably, random pruning also exhibits a non-trivial phase gradient on LLaMA (Wilcoxon $p <$ []), reducing the contrast between strategies. This dissociation between qualitatively consistent direc-

tional trends and markedly different magnitudes suggests the phase-strategy interaction may be a general property of compound sparsity on reasoning models, while its quantitative impact—and the degree to which it is strategy-specific—depends on architecture-specific attention distributions and head specialization patterns.

8 Conclusion

We showed that compound sparsity severity—the super-additive interaction between head pruning and KV cache eviction—is not phase-invariant but exhibits an emergent phase-strategy interaction spanning over an order of magnitude within a single generation. Under importance-based eviction, the severity gradient grows from $69\times$ to $274\times$ across sparsity levels (d_z: $0.44 \rightarrow 0.80$); random eviction shows no such modulation (7/8 conditions non-significant). A crossover at s=0.25—where importance-based eviction reverses ranking across phases—demonstrates that no single strategy dominates uniformly. The pattern replicates on Qwen2.5-7B-Instruct at $s \geq 0.20$ ($d_z \geq 0.55$) and partially extends cross-architecture to DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM) with a qualitatively consistent but substantially weaker trend at $\sim$1/40 the magnitude. Discrimination collapse provides a mechanistic explanation supported by both Qwen2-based models; LLaMA results are directionally consistent but the interaction is weaker than in Qwen2 architectures. These findings motivate phase-aware sparsity allocation for reasoning model deployment.

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Limitations

Model scale and architecture. All experiments are conducted at the 7–8B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Preliminary cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, §5.5) partially confirms the qualitative pattern beyond the Qwen2 architecture—monotonically increasing d_z with within-importance phase gradient $\sim$1.0—but with substantially weaker interaction magnitude ($\sim$1/40 of DeepSeek) and mixed statistical evidence (only s=0.30 significant after Bonferroni correction). The pattern has been tested on two architecture families (Qwen2ForCausalLM, LlamaForCausalLM); comprehensive validation across more diverse architectures (e.g., Mistral, multi-latent attention) and larger scales is needed for full generalization claims.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. Different reasoning tasks—code generation, commonsense reasoning, multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile.

Mechanism. The discrimination collapse mechanism is supported by correlational evidence (consistent Gini/CV reduction across all conditions);

causal validation through controlled entropy manipulation is needed. The entropy redistribution component replicates on only 2/4 Qwen conditions, suggesting model-specific factors.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics. We do not report inference throughput or latency measurements, which are necessary for a complete deployment analysis but orthogonal to our severity characterization. Extended limitations in Appendix I.

A Compound Sparsity Formulation

Attention head pruning. Heads are ranked by Frobenius norm importance $w_j^{(\ell)} = \|\mathbf{A}_j^{(\ell)}\|_F$ (Voita et al., 2019; Michel et al., 2019); a binary mask $m_j^{(\ell)} \in \{0, 1\}$ retains the top-$(1-s) \cdot H$ heads:

$$\tilde{\mathbf{o}}_j^{(\ell)} = m_j^{(\ell)} \cdot \mathbf{o}_j^{(\ell)}. \tag{3}$$

KV cache eviction. A position mask $\mathbf{m}_k \in \{0, 1\}^T$ retains the top-$(1-s) \cdot T$ positions (BOS always preserved):

$$\tilde{a}_{ij} = a_{ij} \cdot m_{k,j}, \quad \forall\, i, j. \tag{4}$$

Under importance-based eviction, each position j is scored by its cumulative attention mass with position normalization:

$$\mathrm{imp}(j) = \frac{\sum_{\ell,h,i} a_{ij}^{(\ell,h)}}{T - j}, \tag{5}$$

where the denominator compensates for causal bias—earlier positions are attended to by more queries (Ramachandran et al., 2026; Feng et al., 2026). Under random eviction, positions (excluding BOS) are evicted uniformly at random.

Compound sparsity. Applying both mechanisms simultaneously yields compound sparsity. The two axes interact: head pruning redistributes attention mass, altering the importance ranking for KV eviction. Prior work (Liao et al., 2026) reports super-additive joint effects but treats interaction magnitude as a fixed property of s.

Attention sparsity metric. For each query position i and head h, we compute the fraction of key positions receiving negligible attention:

$$\sigma_i^{(h)} = \frac{1}{T} \sum_{j=1}^{T} \mathbf{1}\left[a_{ij}^{(h)} < 0.01 \cdot \max_k a_{ik}^{(h)}\right]. \tag{6}$$

The per-position sparsity $\sigma_i = \frac{1}{H}\sum_h \sigma_i^{(h)}$ averages across all heads. Tokens are partitioned into high-, mid-, and low-sparsity phases by the tercile boundaries of $\{\sigma_i\}$; segments shorter than 10 tokens are merged into adjacent phases.

B Multiplicative Null Verification

We verified that replacing the additive null $r_b = r_a + r_k - 1$ (Eq. 1) with the multiplicative null $r_b = r_a \cdot r_k$ does not change qualitative conclusions. Under the multiplicative formulation, the interaction index becomes $\mathrm{II}_\times = r_b - r_a \cdot r_k$. Across all sparsity levels and both eviction strategies, the sign and relative ordering of $\mathrm{II}_\times$ match those of the additive II: importance-based eviction exhibits a monotonically increasing severity gradient ($G_\times = 58$–$241\times$), while random eviction remains flat ($G_\times \approx 1$). The crossover at s=0.25 is preserved. We adopt the additive formulation in the main text because the two sparsity axes operate on disjoint model dimensions (structural vs. positional), making additive independence the more natural null hypothesis at the sparsity levels studied.

C Phase-Strategy Crossover Details

At s=0.25, a strategy ranking reversal emerges (Figure 2): importance-based eviction yields *lower* II than random in the high-sparsity phase (0.032 vs. 0.818) but *higher* II in the low-sparsity phase (7.196 vs. 1.363—a 5.3× reversal). This crossover is present at all four sparsity levels, with reversal magnitude of 3.9× at s=0.15, 5.4× at s=0.20, 5.3× at s=0.25, and 3.6× at s=0.30 (as random eviction severity also rises). The reversal arises because head pruning redistributes attention mass, altering importance rankings in a way that couples the two sparsity mechanisms in low-sparsity phases (§F.1).

D Cross-Model Replication Details

Both models share the Qwen2ForCausalLM backbone (28 layers, 28 heads, 4-group GQA) but differ in training: DeepSeek uses reasoning distillation from DeepSeek-R1, while Qwen2.5 uses standard instruction tuning without explicit reasoning delimiters. This controlled comparison isolates the effect of training methodology on the compound sparsity interaction.

Despite lacking explicit `<think>` delimiters, Qwen2.5-7B-Instruct exhibits sufficient variation

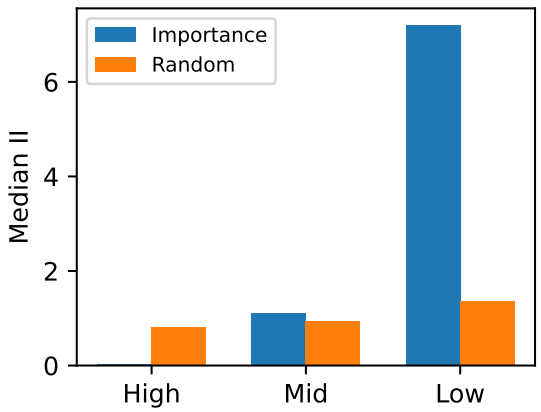


Figure 2: Strategy-dependent crossover at s=0.25. Grouped bar chart of median II by eviction strategy and reasoning phase. Importance-based eviction outperforms random in the high-sparsity phase but produces 5.3× higher II in the low-sparsity phase, reversing the strategy ranking.

Table 5: Per-phase effective sample size for Qwen2.5-7B-Instruct ($n_{\text{total}} = [N_QW]$).

Phase	s=0.15	s=0.20	s=0.25	s=0.30
High	[]	[]	[]	[]
Mid	[]	[]	[]	[]
Low	[]	[]	[]	[]

Table 6: Cross-model comparison of median compound severity difference (Δ) under importance-based eviction.

Model	s=0.15	s=0.20	s=0.25	s=0.30
DeepSeek-R1-Distill-Qwen-7B	0.39	1.48	7.16	14.59
Qwen2.5-7B-Instruct	0.04	0.23	1.27	2.94

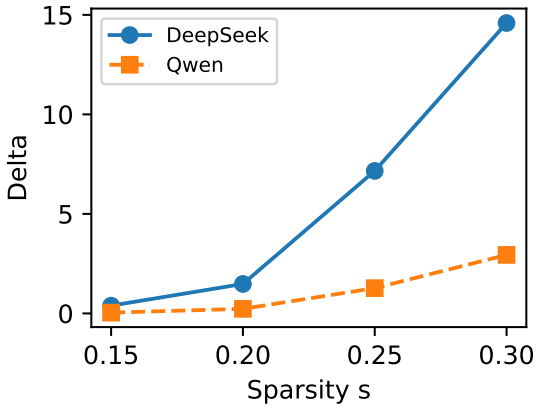


Figure 3: Cross-model comparison of the median severity difference under importance-based eviction. Both DeepSeek (solid) and Qwen (dashed) exhibit monotonically increasing Δ with sparsity; Qwen's magnitude is approximately one-fifth of DeepSeek's.

in attention sparsity to permit phase segmentation via the same quantile-based method (§3). Four of the 50 problems are excluded due to degenerate phase assignments, yielding n=46 valid problems for Qwen (vs. n=49 for DeepSeek).

Per-phase effective sample size. Table 5 reports the per-phase effective sample size for Qwen2.5-7B-Instruct, defined as the number of problems with $|\text{II}| > 0.001$ in each phase. The mid-phase null rate ([NULL_PCT]%) reflects problems where compound interaction is absent in intermediate phases; this does not affect the low–high contrast (Δ) used in the primary analysis.

Table 6 reports the cross-model comparison. Qwen reproduces the monotonic severity gradient at ∼1/5 of DeepSeek's magnitude. Figure 3 visualizes this comparison.

Per-problem concordance. The per-problem concordance across models is not significant (median ρ = [], p = []), indicating that while the aggregate gradient direction replicates, the specific problems driving severity differ between models—consistent with model-specific attention patterns rather than a universal per-problem vulnerability ranking.

Random eviction control. As a control, random eviction on Qwen yields no systematic phase gradient at three of four sparsity levels (3/4 FAIL), mirroring DeepSeek's 4/4 FAIL pattern. At s=0.30, Qwen random produces a borderline PASS (permutation p = 0.012) but with negligible median difference ($\widetilde{\Delta}$ = [], CI spans zero), indicating no meaningful phase modulation.

E Permutation Test Details

For each problem i and sparsity level s, we compute the compound severity difference $\Delta_i^{(\text{strat})} = \text{II}_{low,i} - \text{II}_{high,i}$ under each eviction strategy. The test statistic is the median of the paired difference $\widetilde{\Delta} = \text{median}_i\big(\Delta_i^{(\text{imp})} - \Delta_i^{(\text{rand})}\big)$, measuring whether importance-based eviction produces systematically larger phase-dependent severity than random eviction. We permute strategy labels 10,000 times to construct the null distribution and apply Bonferroni correction for 4 sparsity-level comparisons.

Per-problem distributions of the paired severity

Table 7: Paired permutation test for compound severity: importance vs. random eviction on DeepSeek-R1-Distill-Qwen-7B (n=49, 10,000 permutations, Bonferroni-corrected).

s	Observed $\widetilde{\Delta}$	95% CI	p_{Bonf}
0.15	0.37	[0.19, 0.62]	0.0032
0.20	0.97	[0.69, 1.49]	<0.001
0.25	3.34	[1.66, 6.64]	<0.001
0.30	9.85	[6.70, 16.29]	<0.001

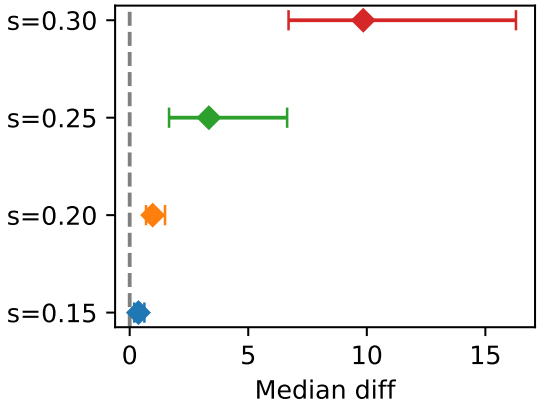


Figure 4: Forest plot of permutation test results. Horizontal bars show 95% permutation CIs; diamonds mark observed median differences $\widetilde{\Delta}$. All CIs lie entirely above zero, confirming significant strategy-dependent severity at every sparsity level.

difference $\Delta_i = \text{II}_{low,i} - \text{II}_{high,i}$ under importance-based eviction are right-skewed at all sparsity levels, with mean/median ratios ranging from 1.2× (s=0.15) to 1.6× (s=0.30) for DeepSeek and 1.1× to 1.4× for Qwen.

The permutation null distributions are approximately symmetric around zero at all sparsity levels. At s=0.15, the observed statistic falls at the 99.92th percentile of the null ($p_{\text{Bonf}} = 0.0032$); at $s \geq 0.20$, no permutation exceeds the observed statistic.

Table 7 reports the results. All four sparsity levels reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30. The monotonic growth of both the point estimate and the CI lower bound ($0.19 \to 0.69 \to 1.66 \to 6.70$) confirms that compound severity amplification strengthens with sparsity—not merely in central tendency but also in worst-case lower bounds. Figure 4 presents these results as a forest plot.

Table 8: Discrimination collapse under importance-based eviction. ΔGini and ΔCV are post-pruning minus baseline; negative values indicate reduced discriminative power.

Model	s	ΔGini	ΔCV
DeepSeek	0.15	−0.030	−0.56
DeepSeek	0.20	−0.040	−0.77
DeepSeek	0.25	−0.065	−1.29
DeepSeek	0.30	−0.081	−1.66
Qwen	0.15	−0.010	−0.05
Qwen	0.20	−0.012	−0.21
Qwen	0.25	−0.010	−1.49
Qwen	0.30	−0.020	−1.14

F Analysis: Entropy-Discrimination Collapse

F.1 Discrimination Collapse

Head pruning reduces the number of heads contributing to cumulative importance scores, flattening the score distribution. Table 8 quantifies this effect: the Gini coefficient decreases by 0.030–0.081 (DeepSeek) and 0.010–0.020 (Qwen) after pruning, with consistent negative deltas across all 16 conditions (both models × 4 sparsity levels × 2 strategies). The CV delta is likewise negative in all 8 importance-based conditions. The same pattern holds under random eviction, confirming that discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

When importance scores lose discriminability, thresholding degenerates into noise-sensitive selection—functionally approaching biased random eviction. This is phase-dependent: in low-sparsity phases, attention is already broadly distributed, so pruning-induced flattening has maximum impact; in high-sparsity phases, concentration is preserved. Random eviction is immune because its decisions are independent of score quality, explaining $G \approx 1$.

F.2 Entropy Redistribution

Beyond discrimination collapse, we examine whether head pruning also redistributes the attention entropy of surviving heads. We measure the mean attention entropy per head:

$$\mathcal{H}_i^{(h)} = -\sum_{j=1}^{T} a_{ij}^{(h)} \log a_{ij}^{(h)} \tag{7}$$

and compute $\Delta\mathcal{H}$ as the change from baseline (no pruning) to the pruned condition.

Table 9: Attention entropy deltas ($\Delta\mathcal{H}$) under importance-based eviction. DeepSeek shows consistent entropy compression; Qwen partially replicates with a reversal at s=0.25.

Model	s	$\Delta\mathcal{H}$
DeepSeek	0.15	-0.040
DeepSeek	0.20	-0.047
DeepSeek	0.25	-0.045
DeepSeek	0.30	-0.044
Qwen	0.15	-0.073
Qwen	0.20	-0.076
Qwen	0.25	$+0.045$
Qwen	0.30	-0.007

For DeepSeek under importance-based eviction, the entropy delta is consistently negative across all four sparsity levels ($\Delta\mathcal{H} \in [-0.047, -0.040]$; Table 9), indicating that importance-based pruning systematically selects heads whose removal concentrates attention in the surviving heads, reinforcing discrimination collapse.

For Qwen, the entropy pattern only partially replicates. The delta is negative at s=0.15 ($\Delta\mathcal{H} = -0.073$) and s=0.20 (-0.076), consistent with DeepSeek. However, at s=0.25, the delta reverses to $+0.045$, with 22 of 30 problems showing positive entropy change. At s=0.30, the delta is near zero (-0.007). The s=0.25 reversal is systematic rather than a statistical artifact: it is consistent across all three reasoning phases (low: $+0.042$, mid: $+0.034$, high: $+0.017$). We interpret this as a genuine model-specific difference, possibly arising from distinct head specialization patterns: Qwen's instruction tuning (vs. DeepSeek's distillation-based training) may produce head configurations where removing 25% of heads triggers compensatory broadening rather than concentration.

We characterize discrimination collapse as the robust, model-independent driver of the phase-strategy interaction, with entropy redistribution as a secondary amplifier that is consistent in DeepSeek but model-dependent in Qwen.

G Discrimination Collapse Details

This cross-model consistency (Table 8) establishes discrimination collapse as a robust consequence of compound sparsity. When importance scores lose discriminative power, thresholding the bottom-s fraction becomes noise-sensitive: small perturbations change which positions cross the eviction boundary. In the limit of perfectly uniform scores, importance-based eviction degenerates into an arbitrary selection functionally equivalent to biased random eviction. The same pattern holds under random eviction (DeepSeek: -0.031 to -0.073; Qwen: -0.019 to -0.045), confirming that discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

H Extended Discussion

On the predictability of phase-modulated severity. The compound interaction is not uniformly fragile—it is selectively fragile in a phase-dependent manner ($\widetilde{\Pi}_{\text{high}} < 0.06$ at $s \leq 0.30$; at $s = 0.35$, high-phase II rises to 0.105 in a supplementary experiment, indicating onset of compound interaction even in high-sparsity phases at extreme pruning rates).

Random masking as falsification control. Random eviction fails 7/8 paired permutation tests across both models; the single exception (Qwen s=0.30, p=0.012) exhibits negligible effect size ($\widetilde{\Delta} = []$). The absence of a phase gradient under random masking isolates the amplification as strategy-specific.

Entropy mechanism boundary conditions. Directly measuring per-head entropy shifts across architecturally distinct models—and identifying what training-time properties determine whether entropy concentrates or disperses under head pruning—would clarify the boundary conditions of the full mechanism.

I Extended Limitations

Model scale. All experiments are conducted at the 7B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Larger models may exhibit different attention redundancy structures that alter the discrimination collapse dynamics (and the model-dependent entropy component may behave differently at scale).

Architectural scope. Our two models—DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct—share the Qwen2ForCausalLM architecture. The cross-model replication (§5.4) establishes that the findings are not artifacts of a single model's weights, though both models share the same Qwen2ForCausalLM architecture. Preliminary cross-architecture validation

on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, §5.5) shows qualitatively consistent within-importance phase gradients ($d_z \sim 1.0$) but a substantially weaker strategy $\times$ phase interaction than in Qwen2-based architectures; random pruning also exhibits a non-trivial phase gradient on LLaMA, reducing the strategy contrast. This partial confirmation (s=0.30 Wilcoxon p_{Bonf}=0.029; $s \leq 0.25$ non-significant) suggests the qualitative pattern may generalize but quantitative impact is architecture-dependent. Comprehensive cross-architecture validation across more diverse designs (e.g., Mistral, multi-latent attention architectures) and scales remains an important next step.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. Different reasoning tasks—code generation, commonsense reasoning, multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile. Generalizing the phase-strategy interaction to broader reasoning domains is an important next step.

Statistical scope. Bootstrap confidence intervals use B=10,000 resamples (recomputed from initial B=1,000 profiling runs). The paired permutation test yields n=49 paired problems for DeepSeek (1 excluded) and n=46 for Qwen (4 excluded). The severity difference distribution is right-skewed (mean/median ratio $\sim$1.6:1 at s=0.30), motivating our use of medians and nonparametric tests rather than parametric alternatives. All four permutation tests reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$; Table 7), with CI lower bounds increasing from 0.19 to 6.70—confirming the effect direction at every sparsity level. The permutation CIs at higher sparsity are wider (e.g., $[6.70,\ 16.29]$ at s=0.30), reflecting genuine variance in per-problem severity amplification.

A notable exception among the random baselines is the Qwen random condition at s=0.30, which yields a statistically significant result (p=0.012) despite negligible effect size ($\widetilde{\Delta} = []$, bootstrap CI spanning zero). This case illustrates that dichotomous significance testing can flag effects of no practical magnitude—reinforcing the importance of reporting effect sizes and confidence intervals alongside p-values.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$, a moderate range representative of practical deployment targets. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics—the sub-exponential growth pattern suggests the severity gradient may plateau, but this remains unverified. Conversely, the lower boundary at which phase-modulated amplification first emerges is not yet characterized.

No throughput evaluation. We do not report inference throughput or latency, which are necessary for deployment analysis but orthogonal to our severity characterization.